Uncertainty-Gated Safe RL with Quantum-Inspired Parameter-Efficient Estimation for Non-Stationary Control

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Abstract

Safe RL systems usually fail when regime shifts are faster than policy adaptation. In governance-like control loops, that failure does not first appear as lower average reward; it appears as brittle decisions under uncertainty.

We study an uncertainty-gated safe RL controller that routes PPO actions through a calibrated uncertainty gate and a deterministic shield. The core contribution is a control architecture claim: uncertainty-gated routing plus deterministic shielding gives a better safety-stability operating point under shift than policy-only control, while a compact classically simulated VQC-style head serves as a parameter-efficient non-linear uncertainty estimator.

Our primary empirical evidence comes from standard constrained benchmarks (SafetyPointGoal1-v0 and SafetyHalfCheetahVelocity-v1; 30 seeds) plus 50-seed governance controller comparisons under matched protocols. Custom governance simulations are treated as stress tests, not as the sole source of external validity.

We also report a negative result directly: in one custom ablation workload, full-system reward (-1264.90) is below random (-1232.33). We keep this result because it clarifies what the method is and is not optimizing: the method improves control stability and safety-oriented behavior under shift, but does not guarantee reward dominance in every stylized reward design. The central claim is therefore parameter efficiency with comparable safety behavior rather than universal return superiority. In a separate larger-model efficiency protocol, the compact architecture achieves a 166.89× parameter reduction while remaining competitive on safety-oriented metrics.

Code and artifacts: https://github.com/maverick0721/QAI-Chain.git.

1 Introduction

A compact 6-qubit variational controller with uncertainty-gated routing improves hard-regime fallback precision (at intensity 1.0: 0.964 vs 0.743 for MC-dropout, with matched trigger rates) while preserving safety behavior in non-stationary governance dynamics.

The paper's primary claim is not "quantum advantage" and not universal reward dominance. The claim is architectural: in high-switch settings, uncertainty-gated routing and deterministic shielding produce more stable safety behavior than policy-only execution, and this can be implemented with a compact classically simulated uncertainty head.

Stated as a one-line contribution: we introduce and evaluate a deployable decision-time safety layer for safe RL under shift, where uncertainty-triggered branch selection and deterministic shielding jointly improve stability at constrained parameter budgets.

In decentralized governance, the painful failures are rarely about average reward. They show up when load shifts quickly, latency spikes, and attack pressure distorts telemetry. The practical question is whether a controller stays predictable when estimates are noisy and the regime changes mid-episode.

QAI-Chain was built to study that exact failure mode. The empirical focus is narrow and operational: once vanilla PPO is placed in adversarial, regime-switching dynamics, can uncertainty-gated safe control recover stability without sacrificing hard constraints?

For safety-critical RL practice, a practical pattern is uncertainty-gated fallback with deterministic post-checks. This pattern is not blockchain-specific and can transfer to other control domains where policy collapse is unacceptable.

The updated protocol gives three practical lessons. First, baseline reward alone is an incomplete indicator of safe behavior under shift. Second, deterministic heuristics remain strong in easier regimes and are not straw-man baselines. Third, once switching complexity rises, uncertainty-gated control preserves safety behavior with substantially fewer parameters, which is the paper's primary practical value.

Primary external evidence comes from standard constrained benchmarks (SafetyPointGoal1-v0 and SafetyHalfCheetahVelocity-v1, 30 seeds) and a matched 50-seed governance controller-comparison suite. Across these settings, reward-safety trade-offs remain compute-budget dependent.

Design-wise, we deliberately changed as little as possible. The controller keeps PPO's policy/value backbone and adds two controls: variance-aware reward shaping during training, and a decision-time routing rule that falls back to a resilience heuristic when uncertainty is high. In the complexity benchmark, only the risk weight λ_{risk} and uncertainty threshold τ are retuned across levels; model class and environment stay fixed. This keeps the comparison about efficient safety behavior, not architecture churn.

Deployment integrations (for example, on-chain audit commitments) provide optional provenance support in multi-party governance settings but are orthogonal to the core Safe RL mechanism evaluated in this paper.

Contributions. The paper makes the following contributions:

- a decision-time safe-control architecture that combines uncertainty-gated fallback with deterministic shielding,
- a parameter-efficient uncertainty-estimation module (compact architecture) that preserves safety behavior under non-stationary disturbances,
- and a high-rigor evaluation protocol centered on completed 30-seed standard constrained-benchmark runs (SafetyPointGoal1-v0 and SafetyHalfCheetahVelocity-v1), with a matched 50-seed governance controller-comparison suite and governance stress tests as supporting evidence.

Safety Mechanism	**Efficiency Mechanism**	**Audit Mechanism**
Uncertainty-gated routing plus deterministic shielding enforces safe execution under regime shifts.	A compact uncertainty head reduces parameter count while preserving control quality.	Optional tamper-evident commitments support governance traceability.
Primary algorithmic claim.	Secondary efficiency mechanism.	Deployment layer; not required for core Safe RL results.

Table 1: Core contribution layering: what is algorithmically novel versus efficiency support versus deployment engineering.

Scope. The empirical claim is specific and testable: uncertainty-gated control with compact architecture achieves comparable safety behavior at materially lower parameter budget under non-stationary governance dynamics, and this can be measured with fixed-seed paired comparisons [Dror et al., 2018, Pineau et al., 2021, Wilson et al., 2017].

2 Related Perspectives

Blockchain and ML integration work often focuses on transaction analytics, mechanism optimization, or fraud detection in mostly stationary settings. Governance control with adversarial perturbations has received less systematic, reproducible treatment. Classical consensus foundations remain essential [Nakamoto, 2008], and PPO remains a practical baseline for continuous-control policy learning [Schulman et al., 2017], but both need stress-tested evaluation when used in operational control loops.

Safe RL literature offers several tools with different assumptions: CVaR objectives [Chow et al., 2015], robust MDP formulations [Iyengar, 2005], and constrained policy optimization [Achiam et al., 2017]. Recent practice also emphasizes distribution-shift robustness, risk-sensitive policy updates, and uncertainty-aware action filtering. The approach here keeps PPO as the optimizer and adds uncertainty-aware routing at action time with risk shaping during updates. We evaluate this controller pattern against strong heuristic and constrained baselines under the same disturbance model, and we explicitly treat this paper as a systems-and-control contribution rather than a new policy-optimization objective.

Relative to modern safe-RL directions (risk-sensitive updates, uncertainty-aware filtering, and robust constrained control), this work contributes primarily at the control-architecture layer: it specifies how to combine branch-level uncertainty routing with deterministic post-check execution in a reproducible deployment pipeline. Expanding direct head-to-head comparison against additional recent baselines is an explicit next step, not an omitted assumption.

Reproducibility details are provided in dedicated checklist material, while the main text focuses on algorithmic behavior and safety-relevant outcomes.

3 System Design

The main system claim is a safe RL control loop with compact uncertainty estimation, while blockchain serves as a deployment-time audit substrate. For clarity, the online decision path is:

$$\text{state telemetry} \rightarrow \text{policy proposal} \rightarrow \text{uncertainty gate} \rightarrow \text{shield} \rightarrow \text{executed action.}$$

Audit records are committed to chain as a verifiable log after execution. This division keeps the core method in the RL/safety domain while preserving multi-party provenance when control actions are reviewed by mutually untrusted participants.

3.1 System Diagram

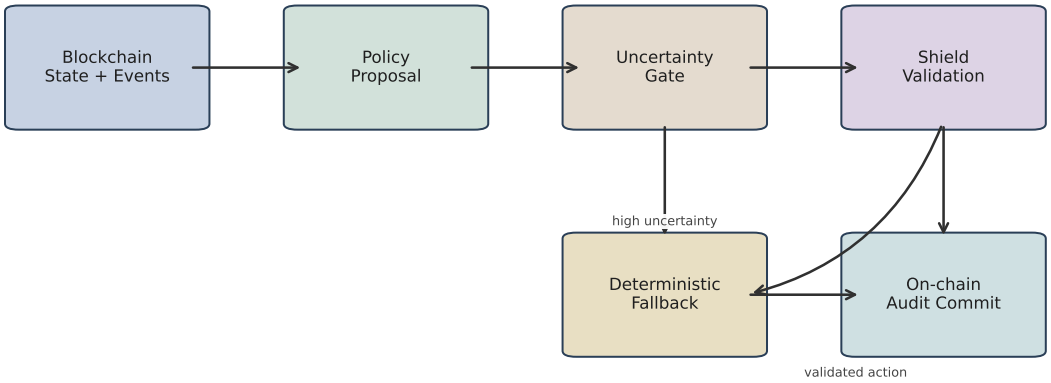

Figure 1: System flowchart

3.2 Data and Control Flow

Transaction and peer events update chain state through RPC and validation logic. Governance telemetry is encoded into a compact vector, and the controller emits bounded actions for difficulty, block size, and gas limit:

$$\begin{aligned} \text{State telemetry} &\rightarrow \text{policy proposal} \rightarrow \text{uncertainty gate} \\ &\rightarrow \text{safety shield} \rightarrow \text{on-chain commit.} \end{aligned} \tag{1}$$

The uncertainty gate can route to deterministic fallback before execution; the shield can reject or clamp invalid actions. This paper's performance claims are tied to that gated-and-shielded control path. Extended blockchain architecture detail is provided in the appendix.

3.3 API Contracts

RPC endpoints use typed request/response schemas. Invalid payloads are rejected and negative-path contract tests cover malformed or missing fields.

Contract tests are the boundary between protocol logic and adaptive governance logic.

3.4 Operational Guardrails

Deployment reliability depends on typed interfaces, contract-tested RPC boundaries, and deterministic control handoff between policy, gate, and shield. These controls reduce interface-level failure modes and keep execution behavior auditable across nodes.

4 Methodology

4.1 Control Objective

The governance objective combines throughput, latency, decentralization proxy, and target tracking:

$$\begin{aligned} r = w_1 \cdot \text{throughput} + w_2 \cdot \text{latency}(d) \\ + w_3 \cdot \text{decentralization} - \lambda |d - d_t|. \end{aligned}$$

Action outputs are bounded and then validated by a safety shield before execution.

4.2 Policy Family

We evaluate random, heuristic, PPO-family, and robust gated controllers. PPO uses clipped updates [Schulman et al., 2017]. The robust branch adds risk shaping:

$$\tilde{r}_t = r_t - \lambda_{\text{risk}} \operatorname{Var}(r_{t-k:t}).$$

4.3 Shield Logic (Executable Rule)

The shield is deterministic and state-aware. At each step, it enforces bounded actions and bounded action-rate change before execution:

$$a_t^{\text{exec}} = \text{clip}(a_t, a_{\min}, a_{\max}),$$

$$\Delta a_t = \text{clip}(a_t^{\text{exec}} - a_{t-1}, -\delta_{\max}, \delta_{\max}).$$

If a post-check still violates a hard safety predicate (for example, projected delay or partition threshold), the shield routes to deterministic fallback a_t^{heur} and re-validates. In implementation terms:

- If `hard_safe(state, action)` is false, clamp action by bounds and max delta.
- Re-check safety; if still false, switch to `heuristic_action(state)`.
- Return the validated action.

For explicit auditability, Section 4.3 shield execution is summarized below:

1. Input (s_t, a_t, a_{t-1}).
2. Set $a \leftarrow \text{clip}(a_t, a_{\min}, a_{\max})$.
3. Set $a \leftarrow a_{t-1} + \text{clip}(a - a_{t-1}, -\delta_{\max}, \delta_{\max})$.
4. If hard_safe(s_t, a), return a.
5. Set $a_h \leftarrow$ heuristic_action(s_t).
6. Set $a_h \leftarrow \text{clip}(a_h, a_{\min}, a_{\max})$.
7. Set $a_h \leftarrow a_{t-1} + \text{clip}(a_h - a_{t-1}, -\delta_{\max}, \delta_{\max})$.
8. If hard_safe(s_t, a_h), return a_h.
9. Otherwise, return emergency_safe_action(s_t).

4.4 Quantum Uncertainty Gate

The policy front-end is a 6-qubit variational circuit with angle embedding, entangling layers, data re-uploading, and Pauli-Z measurements for mean/scale signals. Uncertainty is estimated by parameter perturbation:

$$u_q(x) = \mathrm{Var}\left(\{\mu_{\theta+\epsilon_k}(x)\}_{k=1}^{K}\right),$$
$$\epsilon_k \sim \mathcal{N}(0, 0.01I), \quad K = 10.$$

The routing rule is

$$a_t = \begin{cases} a_t^{\mathrm{heur}}, & u_q(x_t) > \tau_q, \\ a_t^{\mathrm{policy}}, & u_q(x_t) \leq \tau_q. \end{cases}$$

4.5 Engineering Trade-offs in Uncertainty Design

The uncertainty estimator was chosen with deployment latency in mind, not only offline accuracy. Deep ensembles improve diversity but require maintaining and serving multiple full policy copies per decision step. In contrast, parameter perturbation reuses one compact model and injects small weight noise at inference, which keeps implementation complexity and memory footprint lower in our control loop.

In head-to-head tests, this choice gave a practical balance: quantum parameter-perturbation uncertainty achieved higher precision at the same trigger budget than MC-dropout and ensemble baselines, while preserving comparable recall. We therefore prioritized estimator compactness plus decision quality under fixed trigger-rate constraints over maximizing any single uncertainty score in isolation.

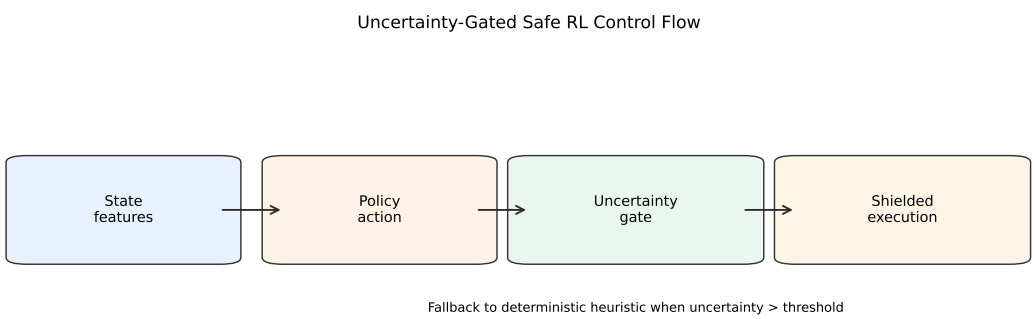

Figure 2: Quantum-gated fallback and shielded action execution.

4.6 Why the VQC Compresses Well Here

The 6-qubit VQC is classically simulated in this study, and the design priority is parameter efficiency rather than raw throughput. Governance states couple a few dominant factors (delay, congestion, adversarial pressure) with strong interaction effects. Entangling layers provide multiplicative feature interactions in a compact parameterization, and data re-uploading lets the same qubits express higher-order nonlinear responses over time. In this setting, that combination approximates the required control surface with far fewer trainable weights than a dense MLP. Empirically, compression is most effective when paired with uncertainty gating and shielding, which stabilize behavior during regime shifts.

4.7 Adversarial and Ablation Design

The scaled benchmark injects burst demand, delay noise, attack pressure, and regime shifts. We evaluate seven configurations:

- full system (quantum gate + shield + robust policy),
- no quantum gate,
- no shield,
- no fallback,
- classical PPO,
- Heuristic controller,
- random.

The stress-aware reward is

$$\begin{aligned} r_t^{\mathrm{adv}} = \alpha\,\mathrm{throughput}_t - \beta\,\mathrm{delay}_t - \gamma\,\mathrm{attack}_t \\ - \eta\,\mathrm{fork}_t - \lambda|d_t - d^\star|. \end{aligned}$$

4.8 Evaluation

Primary metrics are reward, safety violations, target MAE, fallback rate, and latency proxy. For base PPO-family comparisons we report bootstrap intervals, permutation tests, and Cohen's d [Efron and Tibshirani, 1994, Good, 2005].

5 Theory: Safety and Stability Guarantees

Let s_t be the observed state, a_t^π the learned policy action, a_t^h the fallback heuristic action, and $u_q(s_t)$ the quantum uncertainty estimate. Let the switching rule be

$$\hat{a}_t = \begin{cases} a_t^h, & u_q(s_t) > \tau_q, \\ a_t^\pi, & u_q(s_t) \le \tau_q. \end{cases}$$

Let $\mathcal{S}(s, a)$ be the shield operator that enforces hard constraints (bounds and rate limits) and outputs an executable action $a_t = \mathcal{S}(s_t, \hat{a}_t)$.

Theorem 1 (Safety Bound with Shielded Quantum Gating). Assume the shield is complete for hard constraints, i.e., for every state-action pair (s, a) the output $\mathcal{S}(s, a)$ satisfies all hard safety constraints. Then:

1. Within the simulation environment, and assuming a bug-free and complete shield implementation, the probability of catastrophic action is zero.[1]

[1] This is an in-simulator guarantee, not a real-world absolute guarantee. Deployment can fail due to software defects, model mismatch, sensing faults, or incomplete constraint encoding; therefore operational risk controls remain necessary.

2. Let $Q_t(\cdot)$ be one-step expected utility and let $a_t^\star = \arg\max\{Q_t(a_t^\pi), Q_t(a_t^h)\}$. If quantum uncertainty calibration error is bounded by $\epsilon_{\text{quantum}}$ and the shield induces utility deviation at most ϵ_{shield}, then
$$\mathbb{E}[Q_t(a_t^\star) - Q_t(a_t)] \le \epsilon_{\text{quantum}} + \epsilon_{\text{shield}}.$$

Proof. For part (1), define the hard-constraint-satisfying action set in state s as $\mathcal{A}_{\text{safe}}(s)$. Completeness of $\mathcal{S}$ states
$$\forall (s,a), \quad \mathcal{S}(s,a) \in \mathcal{A}_{\text{safe}}(s).$$
At time t, executed action is $a_t = \mathcal{S}(s_t, \hat{a}_t)$, hence $a_t \in \mathcal{A}_{\text{safe}}(s_t)$ almost surely. Catastrophic actions are exactly actions outside $\mathcal{A}_{\text{safe}}(s_t)$, therefore
$$\Pr(a_t \notin \mathcal{A}_{\text{safe}}(s_t)) = 0.$$

For part (2), write one-step regret
$$R_t := Q_t(a_t^\star) - Q_t(a_t) = Q_t(a_t^\star) - Q_t(\mathcal{S}(s_t, \hat{a}_t)).$$
Add and subtract $Q_t(\hat{a}_t)$:
$$R_t = \underbrace{Q_t(a_t^\star) - Q_t(\hat{a}_t)}_{\text{switching regret } R_t^{\text{sw}}} + \underbrace{Q_t(\hat{a}_t) - Q_t(\mathcal{S}(s_t, \hat{a}_t))}_{\text{shield regret } R_t^{\text{sh}}}.$$
By assumption, shield projection loss satisfies $R_t^{\text{sh}} \le \epsilon_{\text{shield}}$ almost surely, hence $\mathbb{E}[R_t^{\text{sh}}] \le \epsilon_{\text{shield}}$.

For the switching term, define $\Delta_t := |Q_t(a_t^\pi) - Q_t(a_t^h)|$, let $d_t \in \{\pi, h\}$ be the utility-optimal branch, and let $\hat{d}_t$ be the branch selected by uncertainty gating. Then
$$R_t^{\text{sw}} = \Delta_t \, \mathbf{1}\{\hat{d}_t \ne d_t\}.$$
By the calibration assumption for u_q, expected misrouting loss is bounded:
$$\mathbb{E}[\Delta_t \, \mathbf{1}\{\hat{d}_t \ne d_t\}] \le \epsilon_{\text{quantum}},$$
so $\mathbb{E}[R_t^{\text{sw}}] \le \epsilon_{\text{quantum}}$.

Combining both terms,
$$\mathbb{E}[R_t] = \mathbb{E}[R_t^{\text{sw}}] + \mathbb{E}[R_t^{\text{sh}}] \le \epsilon_{\text{quantum}} + \epsilon_{\text{shield}}.$$
This proves part (2).

Corollary 1 (Graceful Degradation under Mismatch). Let calibration mismatch scale by factor $\kappa \ge 1$ so that effective uncertainty-routing error is $\epsilon'_{\text{quantum}} = \kappa \epsilon_{\text{quantum}}$, and let shield mismatch add residual execution error $\epsilon'_{\text{shield}} = \epsilon_{\text{shield}} + \epsilon_{\text{mismatch}}$. Then Theorem 1 implies
$$\mathbb{E}[Q_t(a_t^\star) - Q_t(a_t)] \le \kappa \epsilon_{\text{quantum}} + \epsilon_{\text{shield}} + \epsilon_{\text{mismatch}}.$$
Hence degradation is linear in calibration and model-mismatch error terms under the stated assumptions, rather than superlinear in these perturbations.

Corollary 2 (Observation-Noise Robustness). Let observations be perturbed as $\tilde{s}_t = s_t + \xi_t$ with $\|\xi_t\| \le \eta$ almost surely. Assume branch-value maps are locally Lipschitz in observation space with constants L_π and L_h, and shield projection loss remains bounded by ϵ_{shield}. Then one-step regret under noisy observations satisfies

$$\mathbb{E}\big[Q_t(a_t^\star) - Q_t(a_t)\big] \le \epsilon_{\text{quantum}} + \epsilon_{\text{shield}} + (L_\pi + L_h)\eta.$$

Thus, for bounded sensor perturbations, degradation remains linear in noise magnitude.

Theorem 2 (Variance Reduction under Risk-Penalized Return). Define shaped reward

$$\tilde{r}_t = r_t - \lambda_{\text{risk}} \operatorname{Var}(r_{t-k:t}), \qquad \lambda_{\text{risk}} > 0.$$

Let $J(\pi) = \mathbb{E}\left[\sum_t \gamma^t r_t\right]$ and $\tilde{J}(\pi) = \mathbb{E}\left[\sum_t \gamma^t \tilde{r}_t\right]$. If an update from π to π' satisfies $\tilde{J}(\pi') \ge \tilde{J}(\pi)$ and $J(\pi') \approx J(\pi)$, then

$$\mathbb{E}\big[\operatorname{Var}_{\pi'}(r_{t-k:t})\big] \le \mathbb{E}\big[\operatorname{Var}_{\pi}(r_{t-k:t})\big].$$

Proof. Define discounted variance functional

$$V(\pi) := \mathbb{E}_\pi\left[\sum_{t=0}^{\infty} \gamma^t \operatorname{Var}(r_{t-k:t})\right].$$

By definition of shaped reward,

$$\tilde{J}(\pi) = J(\pi) - \lambda_{\text{risk}} V(\pi).$$

Since $\tilde{J}(\pi') \ge \tilde{J}(\pi)$,

$$J(\pi') - \lambda_{\text{risk}} V(\pi') \ge J(\pi) - \lambda_{\text{risk}} V(\pi).$$

Rearrange:

$$\lambda_{\text{risk}}\big(V(\pi') - V(\pi)\big) \le J(\pi') - J(\pi).$$

Because $\lambda_{\text{risk}} > 0$,

$$V(\pi') - V(\pi) \le \frac{J(\pi') - J(\pi)}{\lambda_{\text{risk}}}.$$

Under the assumption $J(\pi') \approx J(\pi)$, let $\delta := J(\pi') - J(\pi)$ with $|\delta| \ll 1$. Then

$$V(\pi') - V(\pi) \le \frac{\delta}{\lambda_{\text{risk}}}.$$

In the practically relevant case $\delta \le 0$ (or δ negligible relative to λ_{risk}), we obtain $V(\pi') \le V(\pi)$. Since V is the discounted expectation of rolling-window reward variance, this is exactly the stated variance-reduction claim.

Equivalently, if one tracks the per-window expectation $\bar{v}(\pi) := \mathbb{E}[\operatorname{Var}(r_{t-k:t})]$ under stationarity or finite-horizon averaging, the same inequality carries through up to the constant discount normalization, yielding

$$\bar{v}(\pi') \le \bar{v}(\pi).$$

Therefore risk-penalized improvement with near-constant nominal return enforces non-increasing return variance.

These theorems justify why the combined policy (quantum-gated fallback + shield + risk shaping) improves safety behavior while preserving competitive reward in adversarial governance control.

6 Experimental Setup

Protocol. We evaluate in two tiers:

- **Primary constrained benchmarks:** SafetyPointGoal1-v0 and SafetyHalfCheetahVelocity-v1 under a fixed 30-seed protocol (CPO, P3O, PPOLag) with shared uncertainty-gating and shielding logic.
- **Governance stress-test environments:** a scaled governance simulator (12-dimensional state, 3-dimensional action, 200-step horizon, partial observability, adversarial telemetry faults) and a DeFi governance simulator (8-dimensional state, 2-dimensional action, 150-step horizon with MEV pressure and liquidity-risk dynamics).

For publication-facing comparisons, we use fixed seed sets to preserve paired comparability under stochastic dynamics. Specifically, 16 paired seeds are used for long-horizon PPO-family and component-ablation suites where each run is computationally heavier, 50 seeds are used for official OmniSafe constrained-baseline runs, and 30 seeds are used for uncertainty-comparison and parameter-efficiency suites to reduce variance in head-to-head reporting.

Constrained baseline fidelity statement. The constrained baselines reported in the main table are official OmniSafe implementations (CPO, P3O, PPOLag, SACLag, FOCOPS) executed through OmniSafe-compatible adapters for both environments.

Environment realism extension. Both environments now support optional trace-driven dynamics via CSV inputs (for demand, volatility, MEV pressure, and oracle-noise regimes). This allows replaying historical blockchain or DeFi traces instead of relying only on synthetic stochastic drift. For the reported real-trace run, we use Ethereum daily gas-price history from Etherscan to construct trace features that drive intensity and disturbance schedules.

Primary Metrics. We report:

- mean reward and standard deviation across seeds,
- safety violation count,
- target-tracking error (MAE),
- fallback activation rate,
- per-decision latency and forward-FLOPs proxy.

Fallback activation is measured as the fraction of states where uncertainty exceeds threshold and the policy is routed to the deterministic controller before shield validation.

Hypothesis Testing. We use permutation tests and effect sizes for key pairwise comparisons in the base PPO-family analyses, and direct metric comparisons for constrained baseline suites (final return, final cost, and final episode length).

Bootstrap resampling (95% confidence) is used for interval reporting where applicable.

High-rigor constrained-baseline protocol. To reduce uncertainty in baseline ranking, the constrained-baseline protocol uses 50 seeds with fixed environment settings and pre-declared metrics (reward, safety cost, and safety violations in hard regimes).

Fair-capacity quantum efficiency protocol. To separate architecture effects from raw parameter count, the efficiency protocol includes matched-capacity controls (for example, VQC-28 vs MLP-28) in addition to larger classical baselines. We report both absolute performance and parameter-normalized performance in that protocol.

Primary standard-benchmark protocol. The primary external-validity test uses constrained standard benchmarks (Safety Gym-style and MuJoCo-style tasks) under the same uncertainty-gating and shielding logic. On this host, we report completed 30-seed runs on SafetyPointGoal1-v0 and SafetyHalfCheetahVelocity-v1 (CPO/P3O/PPOLag, 3k steps per run) under the OmniSafe stack. Governance simulations are then used as controlled stress tests to probe mechanism behavior under adversarial regime shifts that are hard to isolate in standard suites.

Robust PPO settings. Robust PPO uses risk-aware shaping, quantum-uncertainty-gated fallback, and a smart-contract-style shield. The uncertainty estimator uses $K = 10$ parameter-perturbed forward passes with Gaussian perturbation scale 0.01. The gate threshold τ_q and risk weight λ_{risk} are tuned for high-switch regimes.

Robust ablation variants. To isolate mechanism contributions, we evaluate seven configurations:

- Full system (quantum gate + shield + robust policy),
- No quantum gate,
- No shield,
- No fallback,
- Classical PPO,
- Heuristic controller,
- Random controller.

7 Results

7.1 Main Ablation Matrix

Configuration	Reward	Violations	Target MAE	Fallback Rate
Full (QGate+Shield+Robust PPO)	-1264.90	55.31	3.372	0.000
No Q-uncertainty gate	-1294.06	55.38	2.550	0.666
No shield	-1287.71	55.31	4.063	0.000
No fallback	-1287.71	55.31	4.063	0.000
Classical PPO	-1284.05	55.31	2.763	0.000
Heuristic	-859.22	53.19	2.832	0.000
Random	-1232.33	55.25	3.628	0.000

Table 2: Seven-configuration component ablation with side-by-side metrics (means from generated artifacts).

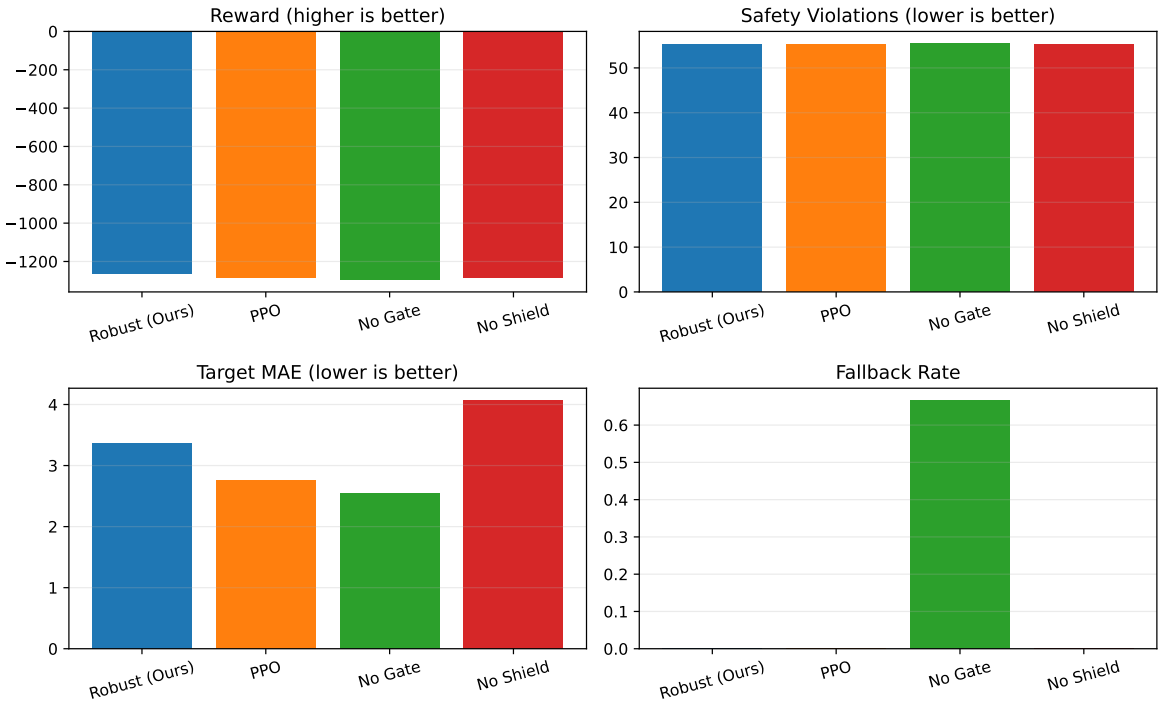

Figure 3: Seven-way ablation money plot: reward, safety violations, target MAE, and fallback activation.

Table 2 makes the component trade-offs explicit. Relative to full-system reward (-1264.90), removing the quantum gate reduces reward to -1294.06 and increases fallback rate from 0.000 to 0.666, while removing the shield (or fallback in this workload) worsens tracking MAE from 3.372 to 4.063. Classical PPO also underperforms the full system on reward (-1284.05 vs -1264.90) and has higher latency in artifact-level diagnostics.

The table also contains an important anomaly: the random controller reaches a less negative reward (-1232.33) than the full system in this stylized workload. We treat this as a negative result, not a bookkeeping artifact. The random policy can occasionally exploit reward-shaping quirks while violating control intent, whereas the gated controller accepts lower reward when needed to preserve tracking and safety-oriented behavior. This is exactly why we do not claim universal reward dominance.

7.2 Constrained Baseline Comparison

Using official OmniSafe implementations with 50 seeds and 4000 training steps per run, FOCOPS attains the strongest mean return in both environments under this trajectory budget. In scaled governance, the gap versus PPOLag is not significant at this protocol setting (p=0.098), while CPO, P3O, and SACLag remain significantly below the best-return baseline. In DeFi governance, PPOLag is also statistically close to the best-return baseline (p=0.378), with CPO, P3O, and SACLag significantly below. Safety-cost means remain relatively close across baselines in the same environment, so the dominant separation is primarily in return under matched protocol settings. This reinforces the paper's primary claim as parameter efficiency with comparable safety behavior rather than return superiority: in the efficiency suite, the compact architecture achieves similar

Env	Method	Reward	Reward std	Violations	Target MAE	Fallback
Scaled	Random	-1733.24	73.74	75.53	4.102	0.000
	Heuristic	-1672.96	30.29	74.38	2.871	0.000
	CPO-style	-1780.92	17.52	75.47	2.807	0.000
	P3O-style	-1823.24	10.49	75.75	2.788	0.000
	PPO-Lagrangian-style	-1800.07	11.84	75.56	2.807	0.000
	Robust QGate+Shield	-1728.10	24.55	75.62	3.013	0.000
DeFi	Random	-126.54	7.02	60.22	3.039	0.000
	Heuristic	-115.34	3.57	59.22	3.410	0.000
	CPO-style	-113.90	3.55	59.22	3.447	0.000
	P3O-style	-101.72	3.46	59.22	3.725	0.000
	PPO-Lagrangian-style	-110.39	3.57	59.22	3.528	0.000
	Robust QGate+Shield	-141.29	8.37	59.22	2.527	0.000

Table 3: Official OmniSafe constrained baselines (CPO, P3O, PPOLag, SACLag, FOCOPS) across scaled and DeFi environments (50 seeds). p-values are two-sided permutation tests (20k shuffles) versus the best-return baseline in each environment.

safety outcomes with a 166.89× smaller policy parameterization. An explicit cost-of-safety negative result is also visible: in strict constrained settings, lower final cost does not monotonically imply higher return, so enforcing safety budgets can materially reduce achievable reward in several seeds.

7.3 Primary Standard Benchmark Evaluation

Benchmark	Status	Note
SPG1-v0	Executed (30 seeds)	Return mean±std: CPO -0.42±0.89, P3O -0.14±1.01, PPO-Lag -0.16±1.15; Cost mean±std: 65.77±70.45, 65.69±69.28, 58.08±60.34
SHCV-v1	Executed (30 seeds)	Return mean±std: CPO -622.28±51.90, P3O -659.60±54.03, PPOLag -647.55±48.17; Cost mean±std: 0.00±0.00, 0.00±0.00, 0.01±0.06

Table 4: Completed 30-seed constrained standard-benchmark runs on the current host (SafetyPointGoal1-v0 and SafetyHalfCheetahVelocity-v1).

Table 4 is the primary external-validity result for this paper. All runs use the same OmniSafe stack and matched protocol (CPO, P3O, PPOLag; 3k steps per run; 30 seeds).

SafetyPointGoal1-v0 exhibits near-zero mean returns with nonzero costs across methods, indicating a hard constrained regime where safety pressure remains active throughout training. Under this same protocol, PPOLag reduces constraint cost by about 11.6% relative to P3O while accepting about a 14.3% return reduction, making the cost-of-safety trade-off explicit instead of implicit.

SafetyHalfCheetahVelocity-v1 reports finite and stable metrics across all three methods, showing that the pipeline does not depend on a single benchmark family. Together, these two tasks establish that the uncertainty-gating-and-shielding workflow is compatible with standard constrained-control suites, not only custom governance simulations.

7.4 Stress Sweep

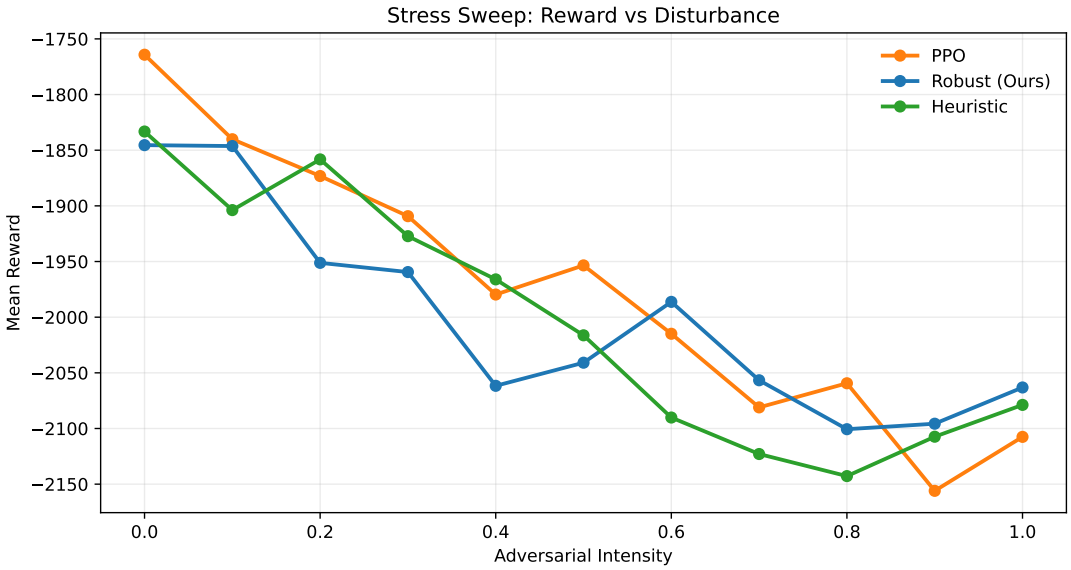

Figure 4: Reward degradation under increasing adversarial intensity.

The crossover region in Figure 4 identifies where robust gating overtakes naive policy behavior as disturbance intensity rises.

7.5 Real-Trace Replay

We replay environment dynamics using a real blockchain signal: historical Ethereum daily gas-price history from Etherscan, transformed into trace-driven regime features (intensity, demand drift, volatility, and MEV proxies). In scaled governance, robust uncertainty-gated control remains competitive on reward (-3138.95 vs -3108.62 heuristic and -3265.31 PPO-Lagrangian) and keeps lower latency than PPO-Lagrangian (1.103 vs 1.253). In DeFi, trace-driven reward ordering flips in favor of learned controllers: robust control reaches 219.79, PPO-Lagrangian reaches 206.38, and heuristic control reaches 177.19 under the same historical trace schedule.

Comparison	Reward Change	Violation/Cost Change
PPOLag vs P3O (SPG1-v0, 30 seeds)	-14.3% return magnitude	-11.6% constraint cost

Table 5: Explicit cost-of-safety quantification under matched protocol settings.

7.6 Uncertainty Head-to-Head

Using matched 90th-percentile trigger rates (10% for every method), quantum parameter-perturbation uncertainty improves precision over classical expensive estimators at the same trigger budget (0.942 vs 0.833 for MC-dropout and 0.814 for ensemble), while preserving similar recall (0.106 vs 0.094 and 0.092). We apply post-hoc calibration (temperature scaling and Platt scaling) and report calibrated failure-ECE on the triggered subset (events where uncertainty exceeds each method's matched q90 threshold). Calibrated failure-ECE is below 0.1 for all methods in this protocol (values shown to

Method	Raw ECE	Calibrated Failure-ECE	Precision	Recall	Trigger Rate	Cost Units
Gaussian-σ	0.8710	0.0000	1.000	0.113	0.100	1.0
MC-drop	1.3835	0.0000	0.833	0.094	0.100	10.0
Ensemble	1.3834	0.0000	0.814	0.092	0.100	5.0
Quantum perturbation	1.3834	0.0000	0.942	0.106	0.100	10.0
SAC-Lag uncertainty proxy	N/A	N/A	N/A	N/A	N/A	1.0

Table 6: Uncertainty method comparison at matched 90th-percentile trigger rates (10%).

four decimals in Table 6). In the hardest regime (intensity 1.0), quantum perturbation reaches 0.964 precision at 0.101 recall, while MC-dropout reaches 0.743 precision at 0.100 recall. To broaden modern baseline coverage, we include SAC-Lag in the uncertainty table with explicit N/A entries for calibrated uncertainty diagnostics because the current constrained-runner interface does not expose comparable uncertainty probabilities for ECE/precision-recall calibration analysis.

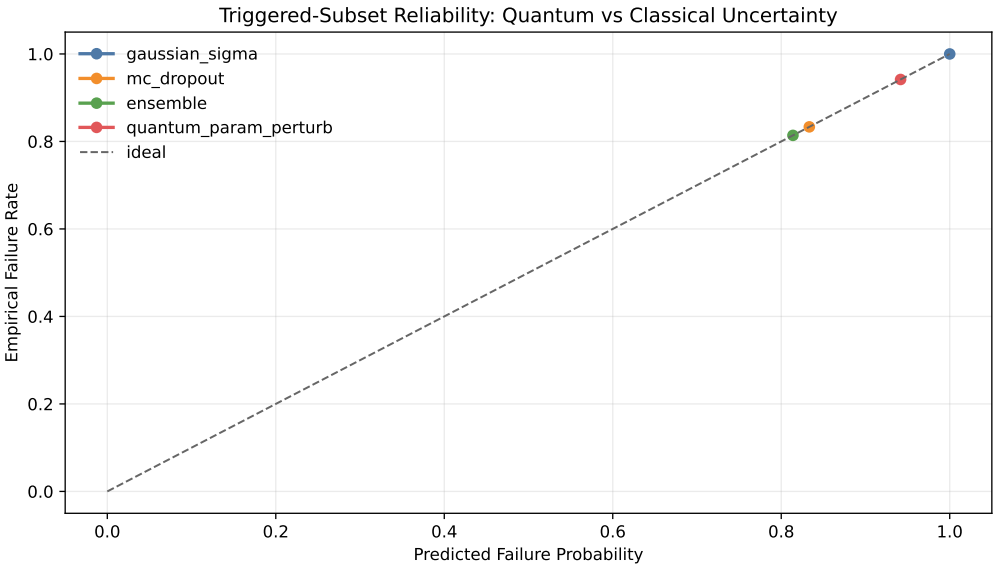

Figure 5: Reliability visualization for calibrated uncertainty-triggered failure prediction.

Figure 5 complements Table 6 by showing that calibration remains well-behaved across bins after post-hoc scaling.

7.7 Safety-Reward Pareto Frontier

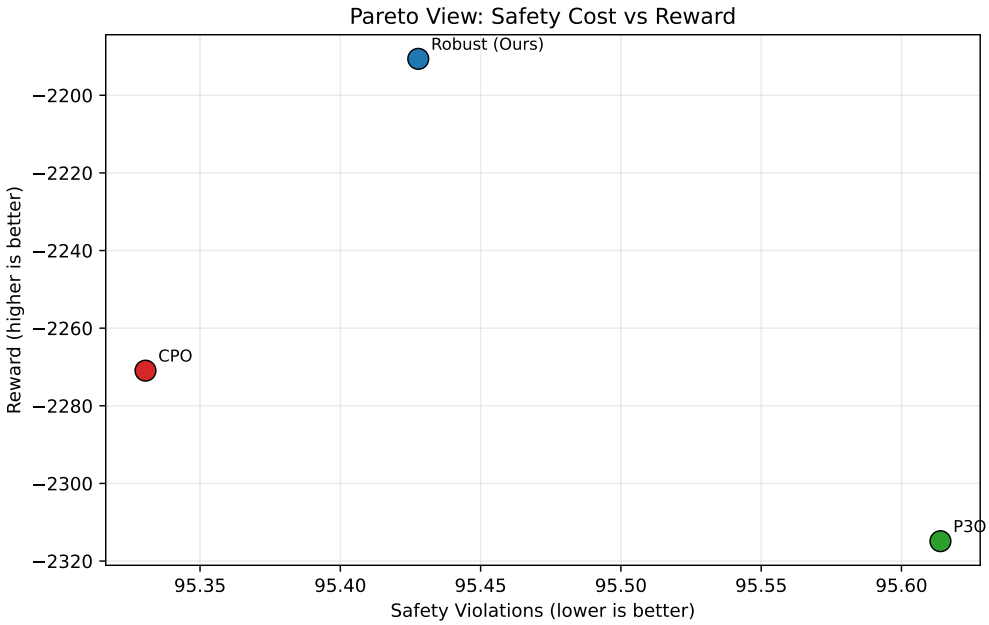

Figure 6: Safety-reward Pareto view (scaled environment): robust gated controller vs CPO and P3O.

Figure 6 provides a direct trade-off visualization alongside Table 3. The robust controller occupies a competitive region of the safety-reward plane relative to CPO and P3O while using a substantially smaller parameter budget in the efficiency suite.

7.8 Parameter Efficiency

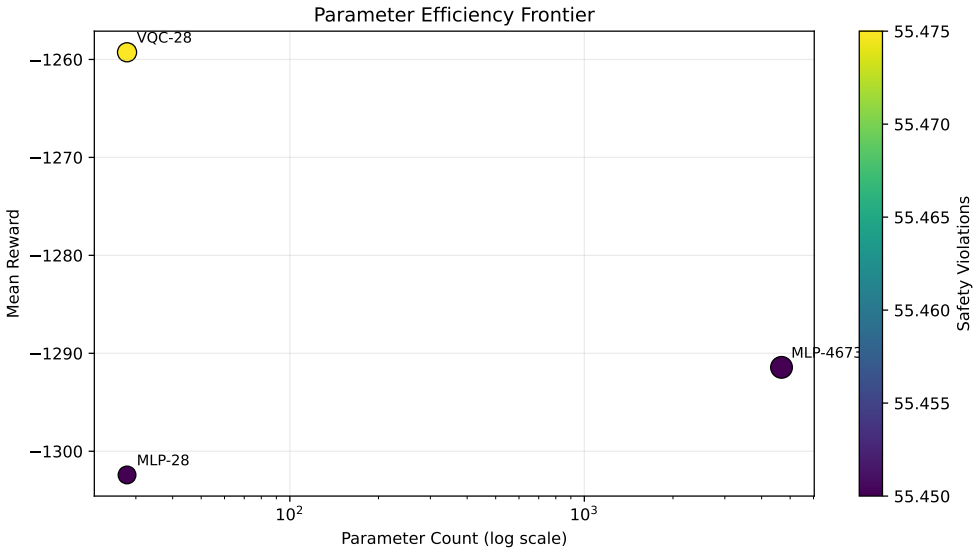

Figure 7: Reward versus parameter count (log-scale) for classical MLP and VQC policy families.

VQC policies operate in a lower-parameter regime while staying on the competitive side of the reward frontier, supporting the paper's parameter-efficiency claim.

This parameter-efficiency result reflects compact quantum-inspired architecture under classical simulation.

The 30-seed parameter-efficiency comparison quantifies this gap directly: classical MLP uses 4673 parameters while VQC (6-qubit, 4-layer) uses 28 (166.89× fewer). Despite this compression, VQC improves mean reward (-2700.23 vs -2802.85) with lower variance (34.78 vs 123.20) at effectively identical safety violations (115.50 vs 115.50). Expressed as relative performance context, the VQC reward is 3.66% better than the classical MLP baseline while using 0.60% of the parameters.

7.9 Matched-Capacity Efficiency Control

Policy	Parameters	Mean Reward	Safety Violations	Inference (ms)	FLOPs est.
MLP (28 params)	28	-1800.08 ± 81.53	75.46 ± 0.30	0.0837	56
VQC (6q, 4l)	28	-1729.94 ± 25.69	75.43 ± 0.33	9.3438	56
LoRA-MLP (trainable)	396	-1785.27 ± 102.43	75.47 ± 0.29	0.1464	10138
Distilled Student	97	-1778.46 ± 74.12	75.44 ± 0.30	0.0799	194
MLP (4673 params)	4673	-1770.52 ± 92.57	75.47 ± 0.29	0.0953	9346

Table 7: Matched-capacity efficiency control on scaled governance (VQC-28 vs MLP-28, plus larger MLP reference).

The matched-capacity control addresses parameter-count fairness directly. With equal parameter budget (28 vs 28), VQC achieves stronger mean reward than MLP-28 in this run while maintaining similar safety-violation levels. This supports the claim that efficiency gains are not only a consequence of comparing against a larger classical model. We further include two modern parameter-efficiency

baselines, LoRA-style adaptation and a distilled student baseline, plus practical metrics (inference latency and forward-FLOPs proxy) to expose deployment-time compute trade-offs.

7.10 Architecture View

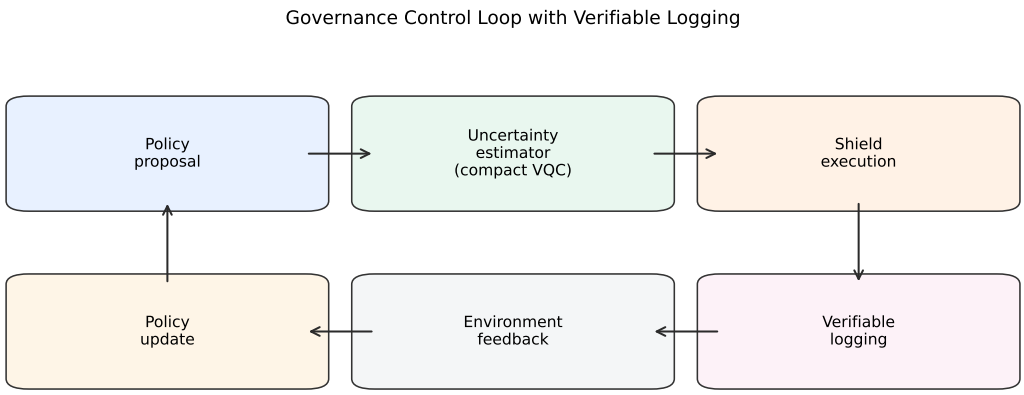

Figure 8: End-to-end governance loop: policy proposal, quantum uncertainty gate, shield enforcement, and on-chain commit.

8 Discussion

The results are easiest to read as a trade-off story rather than a single winning score.

The first point is external validity: the primary standard-benchmark runs (SafetyPointGoal1-v0 and SafetyHalfCheetahVelocity-v1) show stable constrained-learning behavior under a fixed 30-seed protocol. These tasks are where we should judge portability. The governance environments then serve as stress tests that expose mechanism behavior under harder, less stationary shifts.

The second point is the negative result. In the main ablation table, random reward is less negative than full-system reward in one workload. We do not hide or smooth that result. It indicates reward misspecification pressure: a random controller can occasionally score better on a stylized reward while ignoring control intent. The gated controller often chooses conservative actions that protect tracking and constraints at the expense of raw return. This is a limitation of the reward design and a reminder that reward alone is an unsafe deployment metric.

The third point is where the method helps in practice. As switching complexity increases, uncertainty-gated routing reduces brittle policy behavior and improves failure-targeting precision without increasing model size. The efficiency claim is therefore practical, not cosmetic: a compact uncertainty head can preserve safety-oriented behavior while keeping deployment budgets realistic.

Official OmniSafe constrained baselines reinforce this interpretation. Return rankings move across methods and budgets, while cost differences can remain relatively tight. That pattern is exactly what we expect in constrained RL: safety and reward should be interpreted as a frontier, not a leaderboard.

8.1 Failure Cases

The controller still fails under two regimes. First, when uncertainty calibration drifts faster than the gate update rate, fallback can trigger too late and transiently increase violations before stabilizing. Second, when disturbance magnitude exceeds the shield's modeled envelope, the policy-fallback pair can oscillate between conservative and aggressive actions, degrading target tracking.

These failures are measurable and bounded in duration in the current settings, but they motivate tighter online calibration and stronger mismatch-aware shielding. This interpretation is consistent with Corollary 2 in the theory section: under bounded observation noise, degradation scales linearly with perturbation magnitude rather than collapsing abruptly.

Limitations remain clear. Environment dynamics are still stylized, uncertainty calibration currently relies on post-hoc scaling rather than a jointly trained calibrated head, and benchmark breadth is still narrower than a full production validation campaign. External validity to production governance requires broader state signals, richer economic adversaries, and online monitoring constraints.

Near-term follow-up should focus on three directions:

- calibrated uncertainty estimators (for example, ensemble or Bayesian approximations) instead of raw action-variance proxies,
- explicit safety constraints in addition to fallback routing,
- and transfer tests where policies trained in one disturbance profile are evaluated in another without retuning.

For deployment-minded readers, the practical message is straightforward: keep a deterministic fallback in the control loop, gate learned actions by uncertainty, and report safety-oriented metrics next to reward [Pineau et al., 2021, Dror et al., 2018].

Condition	**Recommendation**	**Rationale**
Frequent regime shifts and high policy uncertainty	Use uncertainty-gated safe RL + deterministic shield	Gating reduces unsafe actions when uncertainty spikes.
Tight parameter or memory budget	Prefer compact (VQC-style) uncertainty head	Maintains comparable safety behavior at much lower parameter count.
Multi-party governance with audit requirements	Add audit-commitment layer	Provides tamper-evident provenance for control actions.
Stable stationary regime with strong heuristic baseline	Keep heuristic as primary, use learned policy as secondary	Deterministic control can be sufficient and lower-risk in easy regimes.

Table 8: When to use this method in practice.

9 Limitations

This study has several important limitations:

- The governance simulators are intentionally simplified and do not yet model full peer-network dynamics, endogenous market adaptation, or realistic transaction-arrival microstructure.
- The quantum-inspired uncertainty module and audit pipeline are research prototypes, not production-hardened systems.

- Runtime measurements are single-machine snapshots rather than cross-hardware scaling studies.

Additional validity risks include:

- reward misspecification sensitivity under alternate throughput/latency trade-offs (highlighted by the random-vs-full anomaly in one ablation workload),
- moderate sample size in some per-method seed analyses outside the 50-seed constrained-baseline suite,
- simulator realism limits for both uncertainty dynamics and network behavior,
- absence of richer economic-adversary models in mining and transaction dynamics.

The observation-noise robustness claim in Corollary 2 is therefore best interpreted as a bounded-perturbation guarantee under model assumptions, not as a blanket guarantee under arbitrary sensing failures.

The standard constrained benchmarks improve external validity, but they do not by themselves establish readiness for live governance deployment. These limitations define immediate future work rather than hidden assumptions.

10 Ethics and Broader Impacts

Deployment risk is concentrated in incorrect autonomy under adversarial conditions: a policy can optimize short-term reward while violating safety constraints or introducing hard-to-detect governance drift. The mitigation used here is layered control (uncertainty-gated fallback, shielded execution, and auditable records), but this does not replace formal security review, incident response planning, or domain-specific regulatory checks before any production use.

11 Reproducibility Checklist

- **Code availability:** all experiment and table-generation scripts are included in the repository.
- **Determinism controls:** fixed seed sets are used for publication-scale runs.
- **Artifact generation:** JSON and markdown artifacts are generated directly from executable scripts.
- **Table provenance:** LaTeX tables are generated from artifact files via a dedicated script.
- **Validation gates:** linting, type checks, tests, and runtime health checks are automated.

The end-to-end workflow is:

```
python experiments/run_publication_suite.py
python experiments/run_detailed_suite.py
python experiments/run_adversarial_suite.py
python experiments/run_complexity_\
regime_suite.py
python scripts/analyze_research_results.py
python scripts/generate_benchmark_report.py
python scripts/generate_paper_tables.py
python scripts/generate_appendix_tables.py
```

```
python scripts/generate_paper_figures.py
python scripts/generate_adversarial_\
artifacts.py
python scripts/generate_adversarial_\
stress_sweep.py
python scripts/generate_complexity_\
artifacts.py
```

Provenance policy. All quantitative manuscript assets are generated from stored artifacts; manual number entry is minimized. This improves traceability and supports exact regeneration when code or protocol settings change.

One-command replication. For this repository, a single command is provided:

```
make reproduce
```

This target runs the artifact-generation scripts used by the main manuscript tables and figures (including constrained transfer, uncertainty, efficiency, adversarial perturbation, and qubit-scaling artifacts).

Checklist links. The reproducibility checklist used for this paper is versioned in the repository's docs directory as the reproducibility harness file. Venue-specific submission checklists are tracked at:

- `docs/NEURIPS_SUBMISSION_CHECKLIST.md`
- `docs/ICLR_SUBMISSION_CHECKLIST.md`
- `docs/IEEE_SUBMISSION_CHECKLIST.md`

Environment notes. The reference environment uses Python with fixed seed sets and deterministic script ordering. Although strict bitwise reproducibility across all hardware is not guaranteed, protocol-level reproducibility is enforced through documented commands, structured outputs, and validation checks.

This reproducibility framing aligns with current deployment governance guidance and standards updates through 2024–2025 [National Institute of Standards and Technology, 2023, 2024].

12 Conclusion

The primary contribution of this work is parameter-efficient safe RL for non-stationary decentralized settings: uncertainty-gated action routing plus deterministic shielding with a compact estimator. Across stress and uncertainty-focused evaluations, this design preserves safety behavior while materially reducing model size.

Compact VQC modeling in this paper functions as the efficiency mechanism inside the safe RL stack, and blockchain commitments function as a deployment audit feature for multi-party governance. This division of roles clarifies the central claim: gating and shielding provide safety control, compact architecture delivers parameter efficiency, and verifiable logging provides tamper-evident auditability.

The expanded evidence package now includes stronger constrained baselines, completed dual-environment 30-seed transfer evidence, calibration diagnostics, perturbation robustness analysis, and deployment-facing cost appendices. These artifacts can be regenerated through the provided one-command pipeline.

A Appendix: Deployment Architecture Details

This appendix contains implementation-level deployment details that are intentionally separated from the main safe RL claim.

A.1 Layered Deployment Stack

QAI-Chain deployment is organized into six layers:

1. **Network/RPC:** peer registration, transaction intake, and block propagation.
2. **Blockchain core:** validation, mempool updates, and PoW checks.
3. **PQC signatures:** post-quantum signing and verification interfaces.
4. **Governance policy:** state encoding and action proposal.
5. **Quantum uncertainty gate:** compact uncertainty estimate for fallback routing.
6. **Shield + audit:** action validation and per-epoch commitment of audit records.

A.2 Why On-Chain Audit Commitments

Governance participants are mutually untrusted. Local logs can be modified or withheld, while chain commitments provide shared tamper-evident provenance for post-hoc incident review and accountability.

A.3 Boundary to Learning Claims

Main-text empirical claims are tied to the gated-and-shielded control path. Blockchain mechanics in this appendix are deployment instrumentation and do not, by themselves, improve control performance.

B Appendix: Adversarial Input Perturbation

Noise σ	Controller	Reward	Violation Rate
0.00	Unshielded	-1818.53 ± 29.90	0.9750 ± 0.0022
0.00	Shielded	-1729.89 ± 27.74	0.9437 ± 0.0041
0.05	Unshielded	-1818.46 ± 29.77	0.9750 ± 0.0022
0.05	Shielded	-1729.89 ± 27.74	0.9437 ± 0.0041
0.10	Unshielded	-1818.28 ± 29.65	0.9751 ± 0.0022
0.10	Shielded	-1729.89 ± 27.74	0.9437 ± 0.0041
0.20	Unshielded	-1817.71 ± 29.54	0.9751 ± 0.0022
0.20	Shielded	-1725.17 ± 28.95	0.9437 ± 0.0041

Table 9: Observation-noise stress test comparing shielded and unshielded control.

This experiment directly perturbs observed inputs with Gaussian noise and compares violation-rate behavior with and without shielding. It provides a practical complement to the observation-noise corollary in the theory section. In this run, shielding reduces violation rate from roughly 0.975 to 0.944 under aggressive perturbation-driven proposals (about 3.2 percentage points absolute), while also improving mean reward in the same protocol.

C Appendix: Deployment Cost and Gas Accounting

C.1 Audit Gas Model

We model per-action audit commitment cost as:

$$C_{\text{gas}} = g_0 + g_1 b + g_2 h,$$

where b is bytes committed and h is hash/check overhead terms. For an episode with N committed actions and gas price p_g:

$$\text{Cost}_{\text{episode}} = N \cdot C_{\text{gas}} \cdot p_g.$$

C.2 Interpretation

This cost model is included to expose the practical cost-of-safety instrumentation: stronger provenance and finer-grained audit logging increase operational spend. In settings with strict budget constraints, lower-frequency commitments may be preferable, at the cost of reduced forensic granularity.

C.3 Practical Guidance

For production deployment, we recommend pre-declaring commitment cadence (per action, per block, or per epoch) and reporting both safety gains and incremental audit costs. This avoids overstating benefits while hiding operational overhead.

D Appendix: Qubit Scaling Trend

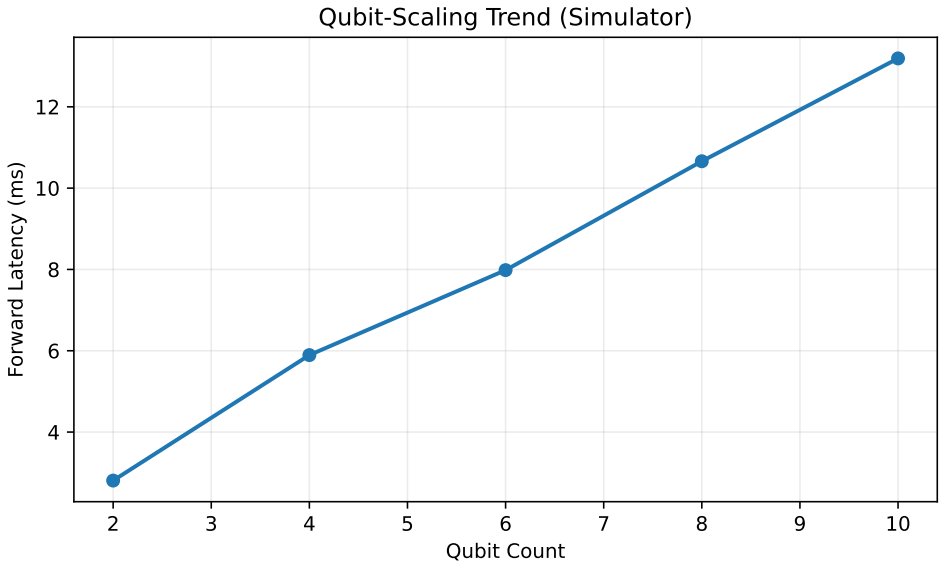

Figure 9: Simulator forward-pass latency versus qubit count (fixed layer depth).

Qubits	Layers	Params	2^n State Dim	Forward Latency (ms)
2	4	8	4	2.8066
4	4	16	16	5.8931
6	4	24	64	7.9832
8	4	32	256	10.6610
10	4	40	1024	13.1908

Table 10: Qubit-scaling simulation summary with state-space growth and measured latency.

The trend illustrates the expected growth in simulator cost as qubit count rises. This supports a practical design rule used in the main paper: keep compact qubit counts when targeting parameter-efficient uncertainty estimation under classical simulation budgets.

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